

Bayesian Learning

Lecture 7 - Gibbs sampling

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Lecture overview

- Monte Carlo simulation
- Gibbs sampling
- Mixture models
- Regularized regression

Monte Carlo sampling

- If $\theta^{(1)}, \dots, \theta^{(m)}$ is an **iid sequence** from $p(\theta|\mathbf{y})$, then

$$\bar{\theta} = \frac{1}{m} \sum_{i=1}^m \theta^{(i)} \rightarrow \mathbb{E}(\theta|\mathbf{y})$$

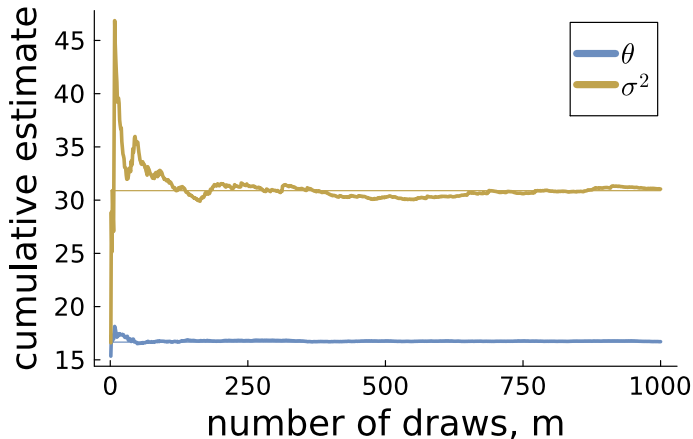
$$\bar{g}(\theta) = \frac{1}{m} \sum_{i=1}^m g(\theta^{(i)}) \rightarrow \mathbb{E}[g(\theta)|\mathbf{y}]$$

for some function $g(\theta)$ of interest.

- **Central limit theorem**

$$\bar{\theta}_{1:m} \stackrel{\text{appr}}{\sim} N\left(\mathbb{E}(\theta|\mathbf{y}), \frac{\mathbb{V}(\theta|\mathbf{y})}{m}\right) \quad \text{for large } m$$

Monte Carlo sampling - convergence



Gibbs sampling

- **Sampling from multivariate distributions**, $p(X_1, \dots, X_p)$.
- Typically a posterior distribution: $p(\theta_1, \dots, \theta_p | \mathbf{y})$.
- Requirement: Easily sampled **full conditional distributions**:
 - ▶ $p(\theta_1 | \theta_2, \theta_3, \dots, \theta_p, \mathbf{y})$
 - ▶ $p(\theta_2 | \theta_1, \theta_3, \dots, \theta_p, \mathbf{y})$
 - ▶ $\vdots$
 - ▶ $p(\theta_p | \theta_1, \theta_2, \dots, \theta_{p-1}, \mathbf{y})$
- Gibbs sampling is a special case of **Metropolis-Hastings**.
- Metropolis-Hastings is a **Markov Chain Monte Carlo (MCMC)** algorithm.

The Gibbs sampling algorithm

Gibbs sampling

Input: initial values $\theta_2^{(0)}, \dots, \theta_p^{(0)}$
number of posterior draws m .

for i in $1:m$ **do**

$$\theta_1 \sim p(\theta_1 \mid \theta_2^{(i-1)}, \theta_3^{(i-1)}, \dots, \theta_p^{(i-1)}, \mathbf{y})$$

$$\theta_2 \sim p(\theta_2 \mid \theta_1^{(i)}, \theta_3^{(i-1)}, \dots, \theta_p^{(i-1)}, \mathbf{y})$$

$$\vdots$$

$$\theta_p \sim p(\theta_p \mid \theta_1^{(i)}, \theta_2^{(i)}, \dots, \theta_{p-1}^{(i)}, \mathbf{y})$$

end

Output: m autocorrelated draws for $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)^\top$
that converge in distribution to the joint
posterior $p(\theta_1, \dots, \theta_p \mid \mathbf{y})$.

Gibbs sampling draws converge to the posterior

- Gibbs draws $\theta^{(1)}, \dots, \theta^{(m)}$ are **dependent**, but

$$\bar{\theta} = \frac{1}{m} \sum_{t=1}^m \theta^{(t)} \rightarrow \mathbb{E}(\theta | \mathbf{y})$$

$$\bar{g}(\theta) = \frac{1}{m} \sum_{t=1}^m g(\theta^{(t)}) \rightarrow \mathbb{E}[g(\theta) | \mathbf{y}]$$

- $\theta^{(1)}, \dots, \theta^{(m)}$ **converges in distribution** to posterior $p(\theta | \mathbf{y})$.
- $\theta_j^{(1)}, \dots, \theta_j^{(m)}$ converges to the marginal posterior of θ_j .
- **Central limit theorem**

$$\bar{\theta} \overset{\text{approx}}{\sim} N(\mathbb{E}(\theta | \mathbf{y}), \mathbb{V}(\bar{\theta})) \text{ for large } m$$

Dependent draws for Gibbs are less efficient

- **Dependent draws** → **less efficient** than iid sampling.
- **IID samples:**

$$\text{Var}(\bar{\theta}) = \frac{\sigma^2}{m}, \quad \text{where } \sigma^2 = \mathbb{V}(\theta|\mathbf{y})$$

- **Autocorrelated samples:**

$$\text{Var}(\bar{\theta}) = \frac{\sigma^2}{m} \left(1 + 2 \sum_{k=1}^{\infty} \rho_k \right)$$

where ρ_k is the autocorrelation at lag k .

- **Inefficiency factor:**

$$\text{IF} = 1 + 2 \sum_{k=1}^{\infty} \rho_k \approx 1 + 2 \sum_{k=1}^K \rho_k$$

- **Effective sample size (ESS):** $\frac{m}{\text{IF}}$.

Gibbs sampling bivariate normal

■ Joint distribution

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \sim N_2 \left[\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right]$$

Gibbs sampling from a bivariate normal

Input: initial value $\theta_2^{(0)}$

number of posterior draws m .

for i in $1:m$ **do**

$$\left| \begin{array}{l} \theta_1^{(i)} \mid \theta_2 \sim N\left(\mu_1 + \rho \frac{\sigma_1}{\sigma_2} (\theta_2^{(i-1)} - \mu_2), \sigma_1^2 (1 - \rho)^2\right) \end{array} \right.$$

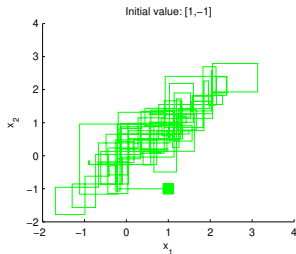
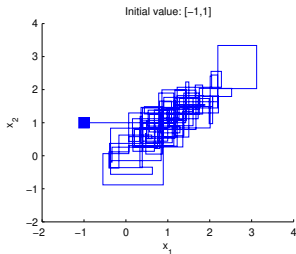
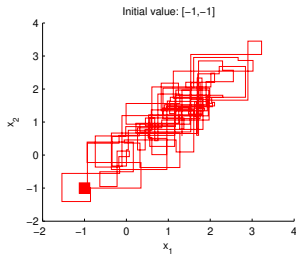
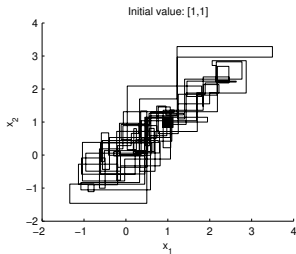
$$\left| \begin{array}{l} \theta_2^{(i)} \mid \theta_1 \sim N\left(\mu_2 + \rho \frac{\sigma_2}{\sigma_1} (\theta_1^{(i)} - \mu_1), \sigma_2^2 (1 - \rho)^2\right) \end{array} \right.$$

end

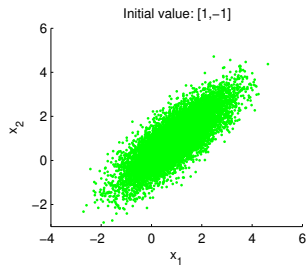
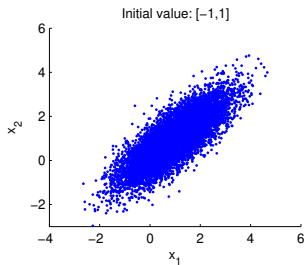
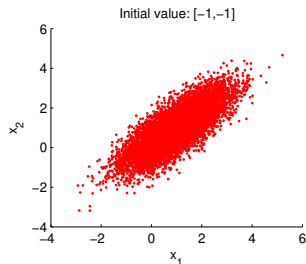
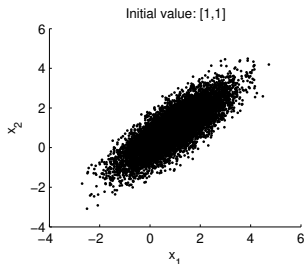
Output: m autocorrelated draws for $\boldsymbol{\theta} = (\theta_1, \theta_2)^\top$ that converge in distribution to the bivariate normal distribution $\boldsymbol{\theta} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$ and

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

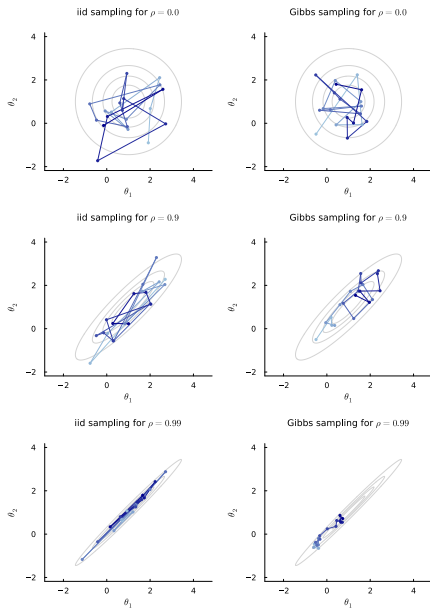
Gibbs sampling - Bivariate normal



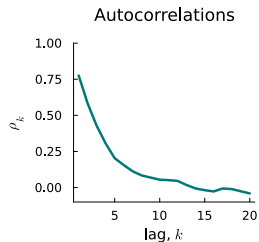
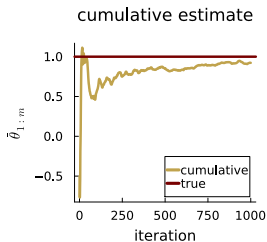
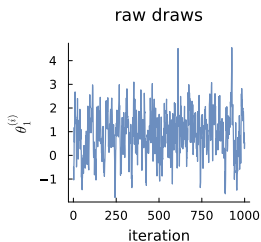
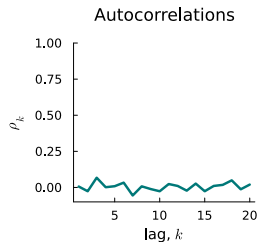
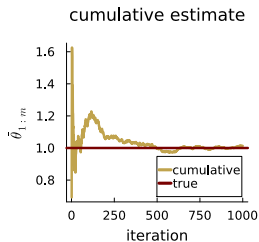
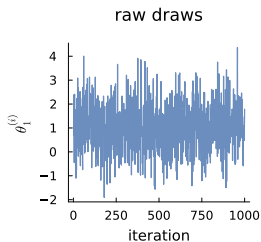
Gibbs sampling - Bivariate normal



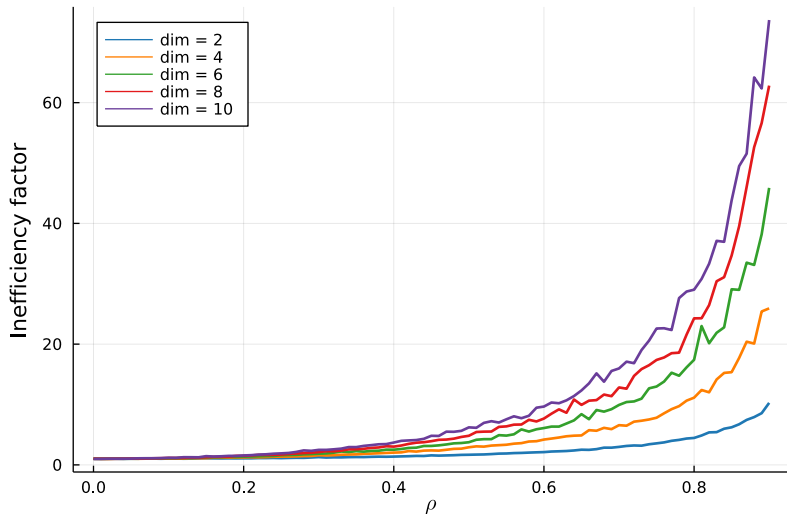
Direct sampling vs Gibbs sampling



Direct vs Gibbs sampling, bivariate normal $\rho = 0.9$



Gibbs is inefficient when parameters are correlated



Normal model with conditionally conjugate prior

■ Normal model with conditionally conjugate prior

$$\begin{aligned}\theta &\sim N(\mu_0, \tau_0^2) \\ \sigma^2 &\sim \text{Inv} - \chi^2(\nu_0, \sigma_0^2)\end{aligned}$$

■ Full conditional posteriors

$$\begin{aligned}\theta | \sigma^2, x &\sim N(\mu_n, \tau_n^2) \\ \sigma^2 | \theta, x &\sim \text{Inv} - \chi^2 \left(\nu_n, \frac{\nu_0 \sigma_0^2 + \sum_{i=1}^n (x_i - \theta)^2}{n + \nu_0} \right)\end{aligned}$$

with μ_n and τ_n^2 defined the same as when σ^2 is known.

Data augmentation - Mixture distributions

■ Let $N(x|\mu, \sigma^2)$ denote the **PDF** of $x \sim N(\mu, \sigma^2)$.

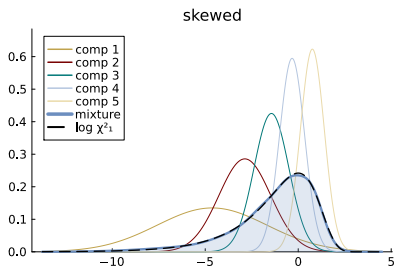
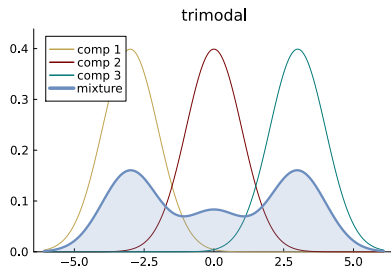
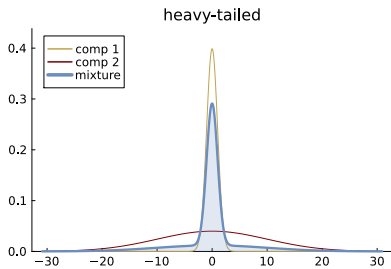
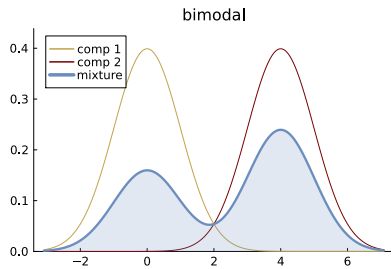
■ Two-component **mixture of normals** [MoN(2)]

$$p(x) = \omega \cdot N(x|\mu_1, \sigma_1^2) + (1 - \omega) \cdot N(x|\mu_2, \sigma_2^2)$$

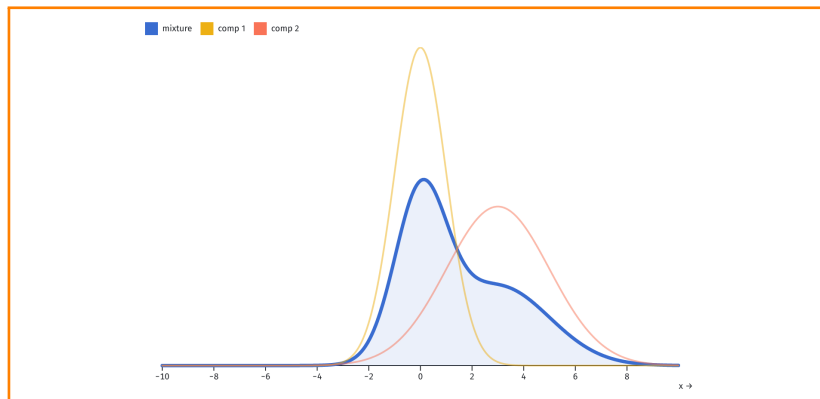
■ **Simulate** from a MoN(2):

- ▶ Simulate a **membership indicator** $Z \in \{1, 2\}$: $Z \sim \text{Bern}(\pi)$.
- ▶ If $Z = 1$, simulate x from $N(\mu_1, \sigma_1^2)$
- ▶ If $Z = 2$, simulate x from $N(\mu_2, \sigma_2^2)$.

Illustration of mixture of normals



Interactive - Mixture of Normals



Data augmentation - Mixture distributions

■ K -component mixture of normals

$$p(x) = \sum_{k=1}^K \omega_k N(x|\mu_k, \sigma_k^2)$$

■ **Indicators:** $Z_i = k$ if observation x_i comes from component k .

Simulating data from a mixture of normals

Input: the number of simulated data observations n

mixture weights $\omega = (\omega_1, \dots, \omega_K)$

mixture component means $\mu_{1:K} = (\mu_1, \dots, \mu_K)$

mixture component variances $\sigma_{1:K}^2 = (\sigma_1^2, \dots, \sigma_K^2)$

for i in $1:n$ **do**

 // Simulate component allocation variable

 Draw $z_i \sim \text{Cat}(\omega_1, \dots, \omega_K)$

 // Simulate from selected mixture component

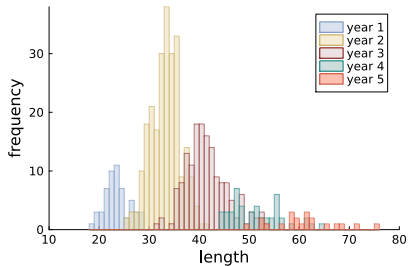
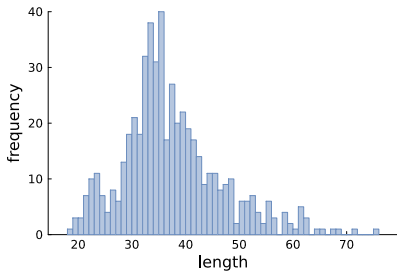
 Draw $x_i|z_i \sim N(\mu_{z_i}, \sigma_{z_i}^2)$

end

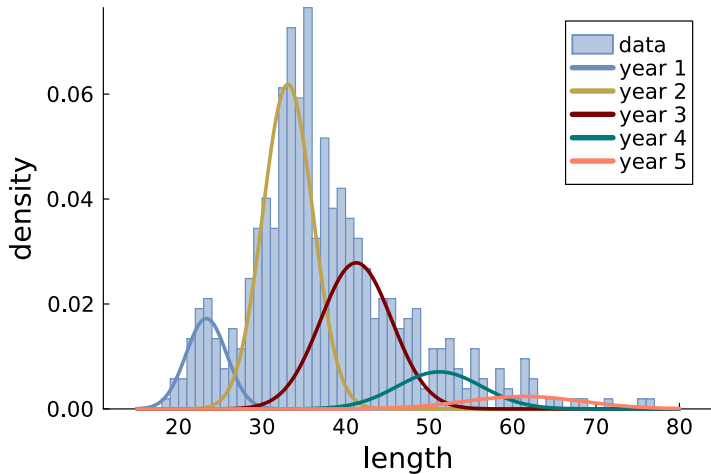
Output: n iid observations $\mathbf{x} = (x_1, \dots, x_n)$ from the mixture of normals model

$$p(x) = \sum_{k=1}^K \omega_k \cdot N(x|\mu_k, \sigma_k^2).$$

Fish length data with known yearly cohorts



Fish length data - fit with known yearly cohorts



Likelihood for a mixture and data augmentation

- The **likelihood** is a product of sums. **Messy** to work with.
- **Assume** that we know where each observation comes from

$z_i = k$ if x_i came from mixture component k .

- **Given** $z_1, \dots, z_n$, **split up the data** in K groups and estimate for each group separately:
 - ▶ means $\mu_1, \dots, \mu_K$
 - ▶ variances $\sigma_1^2, \dots, \sigma_K^2$
 - ▶ mixture proportions $\omega_1, \dots, \omega_K$
- But we do **not** know $z_1, \dots, z_n$!
- **Data augmentation**: add $z_1, \dots, z_n$ as unknown parameters, and update them in separate Gibbs step.

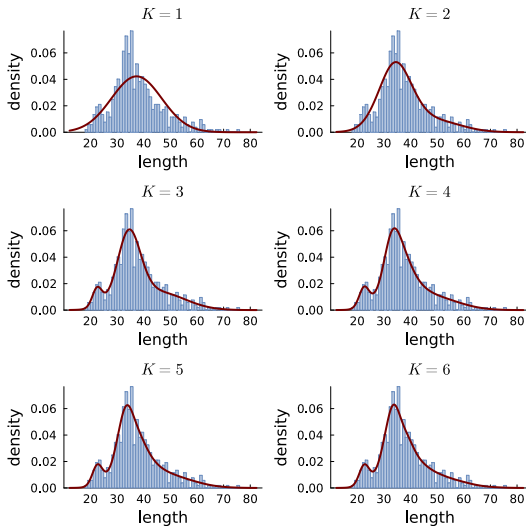
Gibbs sampling for mixture distributions

```
for j in 1:m do
  // Update component parameters
  for k in 1:K do
    Set  $\mathbf{x}_k = \{x_i \text{ such that } z_i^{(j-1)} = k\}$ 
    Draw  $(\sigma_k^2)^{(j)} | \mathbf{x}_k \sim \text{ScaledInv-}\chi^2(\nu_{n,k}, \sigma_{n,k}^2)$ 
    Draw  $\mu_k^{(j)} | (\sigma_k^2)^{(j)}, \mathbf{x}_k \sim N(\mu_{n,k}, \tau_{n,k}^2)$ 
  end

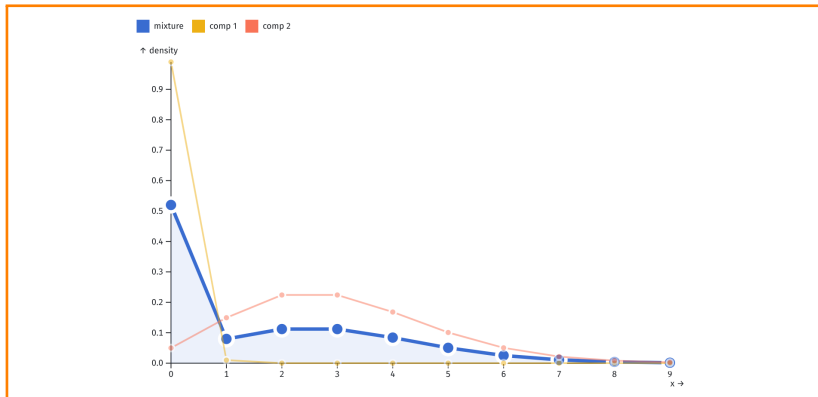
  // Update component weights
  Set  $n_k = |\mathbf{x}_k|$ , number of obs in component k
  Draw  $\omega^{(j)} \sim \text{Dirichlet}(\alpha_1 + n_1, \dots, \alpha_K + n_K)$ 

  // Update mixture allocations
  for i in 1:n do
    for k in 1:K do
       $\tilde{\omega}_k \propto \omega_k^{(j)} \cdot N(x_i | \mu_k^{(j)}, \sigma_k^{(j)})$ 
    end
    normalize  $\tilde{\omega}_1, \dots, \tilde{\omega}_K$  to sum to one
    simulate allocation  $z_i^{(j)} \sim \text{Cat}(\tilde{\omega}_1, \dots, \tilde{\omega}_K)$ 
  end
end
```

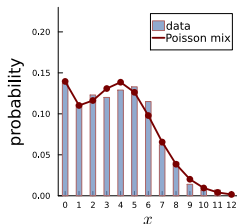
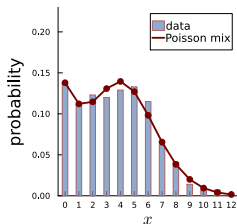
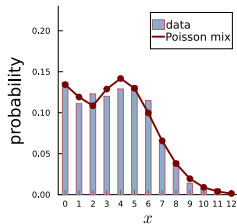
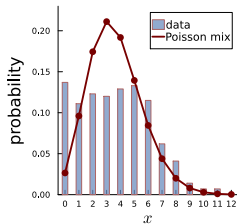
Fish length data - mixture of normals fit



Interactive - Mixture of Poissons



Fitting a mixture Poissons to the eBay bidders



Direct sampling L2-regularized regression

- **Hierarchical** model setup:

$$\mathbf{y}|\boldsymbol{\beta}, \sigma^2, \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$$

$$\boldsymbol{\beta}|\sigma^2, \lambda \sim N(0, \sigma^2 \lambda^{-1} I_m)$$

$$\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$

$$\lambda^{-1} \sim \text{Inv-}\chi^2(\omega_0, \psi_0^2)$$

- Posterior from **conditional-marginal decomposition**

$$p(\boldsymbol{\beta}, \sigma^2, \lambda | \mathbf{y}, \mathbf{X}) = p(\boldsymbol{\beta} | \sigma^2, \lambda, \mathbf{y}, \mathbf{X}) p(\sigma^2 | \lambda, \mathbf{y}, \mathbf{X}) p(\lambda | \mathbf{y}, \mathbf{X})$$

- However, $p(\lambda | \mathbf{y}, \mathbf{X})$ is not a known distribution.

- **Gibbs sampling** can instead be used:

- ▶ Sample $\boldsymbol{\beta} | \sigma^2, \lambda, \mathbf{y}, \mathbf{X}$ from Normal
- ▶ Sample $\sigma^2 | \boldsymbol{\beta}, \lambda, \mathbf{y}, \mathbf{X}$ from $\text{Inv-}\chi^2$
- ▶ Sample $\lambda^{-1} | \boldsymbol{\beta}, \sigma^2, \mathbf{y}, \mathbf{X}$ from $\text{Inv-}\chi^2$

- λ is **easy** to simulate **conditional on** $\boldsymbol{\beta}$ and σ^2 . See book.

Improving the efficiency of the Gibbs sampler

- **Efficient blocking.** Correlated parameters should ideally be included in the same updating block.
- **Reparametrization.** Convergence can improve dramatically in alternative parametrizations.
- **Data augmentation**
 - ▶ make **full conditional posteriors more easily sampled**
 - ▶ examples: mixture models, missing data etc etc
 - ▶ but typically **increases the autocorrelation** between draws